THE HERITAGE SCHOOL
FIRST TERM EXAMINATION, 2026–27
PHYSICS PAPER 1 (THEORY) · CLASS 12 · PRACTICE PAPER 2

Maximum Marks: 70 **Time Allotted: Three Hours**

Instructions to Candidates

- You are allowed an additional fifteen minutes for only reading the question paper. You must NOT start writing during this time.
- There are **twenty questions** in this paper. **Answer all questions.**
- There are **four sections** in the paper: A, B, C and D. Internal choices have been provided in **two questions** each in Sections B, C and D.
- **Section A** consists of one question having fourteen sub-parts of **1 mark** each.
- While attempting the multiple choice sub-parts you are required to write only ONE option as the answer.
- **Section B** consists of seven questions of **2 marks** each.
- **Section C** consists of nine questions of **3 marks** each.
- **Section D** consists of three questions of **5 marks** each.
- The intended marks for questions or parts of questions are given in brackets [ ].
- All working, including rough work, should be done on the same sheet as, and adjacent to, the rest of the answer.
- A list of useful physical constants is given at the end of this paper.
- A simple scientific calculator without a programmable memory may be used for calculations.

Syllabus covered: Electrostatics; Current Electricity; Magnetic Effect of Current and Magnetism; Electromagnetic Induction and Alternating Current; Electromagnetic Waves; Optics (Ray Optics and Optical Instruments, and Wave Optics).

Not examined in this paper: Dual Nature of Radiation and Matter, Atoms and Nuclei, Electronic Devices. Polarisation is no longer in the Wave Optics syllabus and is not examined here either.

SECTION A – 14 MARKS

Question 1 [14 × 1]

In sub-parts (i) to (vii) choose the correct alternative. Answer sub-parts (viii) to (xiv) briefly.

(i) Two point charges $+4q$ and $+q$ are separated by a distance d. The neutral point lies on the line joining them, at a distance from $+4q$ of:

(a) $d/3$
(b) $2d/3$
(c) $d/4$
(d) $3d/4$

(ii) Two bulbs rated 100 W, 220 V and 60 W, 220 V are connected in **series** across a 220 V supply. Then:

(a) the 100 W bulb glows brighter.
(b) the 60 W bulb glows brighter.
(c) both glow equally brightly.
(d) neither bulb glows.

(iii) A proton and an alpha particle, accelerated from rest through the same potential difference, enter a magnetic field at right angles. The ratio of the radii of their paths $r_p : r_\alpha$ is:

(a) 1 : 1
(b) 1 : √2
(c) √2 : 1
(d) 1 : 2

(iv) The self-inductance of a coil is doubled and the rate of change of current through it is halved. The induced emf:
- (a) is doubled.
- (b) is halved.
- (c) remains the same.
- (d) becomes four times as large.

(v) A ray of light passes from glass (n = 1·5) into water (n = 4/3). The critical angle for this pair is nearest to:
- (a) 41·8°
- (b) 48·6°
- (c) 62·7°
- (d) 90°

(vi) **Assertion:** in a single-slit diffraction pattern the central maximum widens when the slit is made narrower. **Reason:** the angular width of the central maximum is $2\lambda/a$, where a is the slit width.
- (a) Both Assertion and Reason are true and Reason is the correct explanation for Assertion.
- (b) Both Assertion and Reason are true but Reason is not the correct explanation for Assertion.
- (c) Assertion is true and Reason is false.
- (d) Assertion is false and Reason is true.

(vii) **Assertion:** a moving coil galvanometer can measure a direct current but not an alternating one. **Reason:** an alternating current would reverse the direction of the torque on the coil every half cycle, so the average deflection would be zero.
- (a) Both Assertion and Reason are true and Reason is the correct explanation for Assertion.
- (b) Both Assertion and Reason are true but Reason is not the correct explanation for Assertion.
- (c) Assertion is true and Reason is false.
- (d) Assertion is false and Reason is true.

(viii) Draw a graph showing the variation of the electric field intensity E with the distance r from the centre of a uniformly charged conducting spherical shell of radius R.

(ix) Define the mobility of a charge carrier and state its SI unit.

(x) Why is soft iron preferred to steel for the core of an electromagnet?

(xi) State Lenz's law.

(xii) Name the electromagnetic radiation emitted by the remote control of a television set.

(xiii) Why does a diamond sparkle?

(xiv) Give a reason why two independent monochromatic sources cannot produce a sustained interference pattern.

SECTION B – 14 MARKS

Question 2 [2]

An electric dipole of moment 4×10^{-9} C m is placed in a uniform field of 5×10^{4} N C^{-1} with its axis at 30° to the field.

(i) Calculate the torque acting on it.

(ii) Calculate the work needed to rotate it until its axis is perpendicular to the field.

Question 3 [2]

A cell of emf 2·0 V and internal resistance 0·50 Ω is connected to an external resistance of 3·5 Ω. Calculate the current and the terminal potential difference.

OR

State any two factors on which the resistance of a given conductor depends, and state how the resistivity of a metal varies with temperature.

Question 4 [2]

A circular coil of 20 turns and radius 8·0 cm carries a current of 5·0 A. Calculate the magnetic flux density at its centre.

Question 5 **[2]**

The current through a coil of self-inductance 2·0 H falls from 4·0 A to 1·0 A in 0·30 s. Calculate the magnitude of the emf induced.

Question 6 **[2]**

State any two characteristics of electromagnetic waves.

OR

Arrange the following in **increasing order of wavelength**: X-rays, microwaves, ultraviolet rays, infrared rays. Which has the greatest frequency?

Question 7 **[2]**

Draw a labelled graph of the angle of deviation δ against the angle of incidence *i* for a prism, and mark the angle of minimum deviation.

Question 8 **[2]**

In a Young's double slit experiment the slits are 0·30 mm apart and the screen is 1·5 m away. If the fringe width is 3·0 mm, calculate the wavelength of the light.

SECTION C – 27 MARKS

Question 9 **[3]**

Derive an expression for the intensity of the electric field at a point on the equatorial (broadside-on) line of an electric dipole of moment *p*, at a distance *r* from its centre. Hence write the result for a short dipole.

Question 10 **[3]**

Draw a labelled circuit diagram of a potentiometer used to determine the internal resistance of a cell. Write the working formula and explain the symbols. (Derivation is not required.)

Question 11 **[3]**

Derive an expression for the torque acting on a rectangular current-carrying loop placed in a uniform magnetic field, and hence define the magnetic dipole moment of the loop.

OR

Using Ampere's circuital law, obtain an expression for the magnetic flux density at a perpendicular distance *r* from a long straight conductor carrying a current *I*.

Question 12 **[3]**

An electron, accelerated from rest through a potential difference of 500 V, enters a uniform magnetic field of $2{\cdot}0 \times 10^{-3}$ T at right angles to its velocity. Calculate the radius of its circular path.

Question 13 **[3]**

Explain the principle and the working of a transformer. Write the relation between the turns and the voltages, and give one reason why the efficiency of a real transformer is never 100 per cent.

Question 14 **[3]**

Using Huygens' wave theory and a neat labelled diagram, prove the law of reflection of light.

Question 15 **[3]**

Starting from the expression for refraction at a single spherical surface, derive the lens maker's formula for a thin biconvex lens.

Question 16 **[3]**

A ray of light falls normally on the face AB of a right-angled isosceles glass prism (45°–90°–45°) of refractive index 1·5. Calculate the critical angle, trace the path of the ray through the prism, and state the total deviation.

OR

An astronomical telescope has an objective of focal length 100 cm and an eyepiece of focal length 5·0 cm. Calculate its magnifying power when the final image is at the least distance of distinct vision.

Question 17 **[3]**

In a single-slit diffraction experiment, state and explain the effect of each of the following.

(i) The slit width is doubled.

(ii) The wavelength of the light is doubled.

(iii) The screen is moved further from the slit.

SECTION D – 15 MARKS

Question 18 **[5]**

(a) A series LCR circuit of a 40 Ω resistor, a 0·40 H inductor and a 40 μF capacitor is connected to a 220 V, 50 Hz supply. Calculate the impedance and the current. **[3]**

(b) Draw a labelled graph showing how the impedance Z of a series LCR circuit varies with the frequency f, marking the resonant frequency. **[2]**

OR

(a) A conducting rod of length l slides with constant velocity v on parallel rails closed by a resistance R, in a field B normal to the circuit. Derive the induced emf, and hence the power an external agent must supply. **[3]**

(b) A rod of length 0·50 m moves at 4·0 m s^{-1} perpendicular to a field of 0·20 T on rails closed by 2·0 Ω. Calculate the induced current and the power dissipated. **[2]**

Question 19 **[5]**

(a) Draw a labelled ray diagram showing image formation by a compound microscope with the final image at the least distance of distinct vision, and write the expression for its magnifying power. **[3]**

(b) The objective and eyepiece of a compound microscope have focal lengths 1·0 cm and 5·0 cm. An object is 1·1 cm from the objective and the final image is at infinity. Calculate the magnifying power. **[2]**

OR

(a) Derive an expression for the magnifying power of a simple microscope when the final image is at the least distance of distinct vision D. **[3]**

(b) What focal length and power of lens would give a magnification of 10 at the near point of 25 cm? **[2]**

Question 20 **[5]**

Read the passage and answer the questions that follow.

A thundercloud and the surface of the earth behave as the two plates of an enormous parallel plate capacitor, with air between them as the dielectric. Charge separation inside the cloud raises the potential difference between its base and the ground to about 10^8 V. When the field becomes large enough to ionise the air, the capacitor discharges in a lightning stroke lasting only about a hundred microseconds. A lightning conductor – a thick copper strip from a spiked rod on the roof to a plate buried in the ground – carries that discharge safely to earth.

(i) Write an expression for the capacitance of a parallel plate capacitor with air between its plates.

(ii) The cloud base has area $4{\cdot}0 \times 10^6$ m^2 and is 2000 m above the ground. Calculate the capacitance.

(iii) Calculate the energy stored when the potential difference is 10^8 V.

(iv) If the discharge lasts 100 μs, calculate the average current in the stroke.

(v) State why the lightning conductor is made of a **thick** copper strip.

USEFUL CONSTANTS AND RELATIONS

	Quantity	Symbol	Value
1.	Permittivity of vacuum	ε_0	$8{\cdot}85 \times 10^{-12}$ F m^{-1}
2.	Constant for Coulomb's law	$1/4\pi\varepsilon_0$	9×10^{9} N m^{2} C^{-2}
3.	Permeability of vacuum	μ_0	$4\pi \times 10^{-7}$ H m^{-1}
4.	Constant in magnetism	$\mu_0/4\pi$	1×10^{-7} H m^{-1}
5.	Charge of a proton = charge of an electron	e	$1{\cdot}6 \times 10^{-19}$ C
6.	Mass of an electron	m_e	$9{\cdot}1 \times 10^{-31}$ kg
7.	Mass of a proton	m_p	$1{\cdot}67 \times 10^{-27}$ kg
8.	Speed of light in vacuum	c	3×10^{8} m s^{-1}
9.	Acceleration due to gravity	g	$9{\cdot}8$ m s^{-2}
10.	Least distance of distinct vision	D	25 cm

ANSWER KEY & MARKING NOTES

Practice Paper 2. **Do not read this section before attempting the paper.** Every numerical answer was recomputed independently; the notes give the decisive step rather than a full solution.

Section A

Q	Answer
1 (i)	**(b)** $2d/3$. Setting $4q/x^2 = q/(d-x)^2$ gives $2/x = 1/(d-x)$, so $x = 2d/3$. Like charges, so the point lies **between** them, nearer the smaller charge.
1 (ii)	**(b)** In series the current is common, so $P = I^2R$ is greatest for the largest R; and $R = V^2/P$ is largest for the **lowest** wattage rating.
1 (iii)	**(b)** $1 : \sqrt{2}$. $r = \sqrt{2mqV}/qB = (1/B)\sqrt{2mV/q}$, so $r \propto \sqrt{m/q}$. For the alpha, m/q is $4m_p/2e = 2m_p/e$, twice the proton's.
1 (iv)	**(c)** $\varepsilon = L(dI/dt)$; doubling one factor and halving the other leaves the product unchanged.
1 (v)	**(c)** $\sin C = n_{water}/n_{glass} = (4/3)/(3/2) = 8/9 = 0{\cdot}889$, so $C = 62{\cdot}7°$.
1 (vi)	**(a)** The angular width varies as $1/a$, so narrowing the slit widens the pattern, and the reason is exactly the cause.
1 (vii)	**(a)** The coil cannot follow mains-frequency reversals, so it simply does not deflect; the reason is the correct explanation.
1 (viii)	$E = 0$ from $r = 0$ to $r = R$; a vertical jump to σ/ε_0 at $r = R$; then falling as $1/r^2$ for $r > R$. Axes must be labelled and the discontinuity at $r = R$ shown.
1 (ix)	Mobility is the magnitude of the drift velocity acquired per unit electric field, $\mu = v_d/E$. SI unit: $\mathbf{m^2\ V^{-1}\ s^{-1}}$.
1 (x)	Soft iron has a **high permeability** and a very **low coercivity and retentivity**, so it magnetises strongly when the current flows and loses almost all its magnetism when the current is switched off – exactly what an electromagnet needs.
1 (xi)	The induced emf, and hence the induced current, is always in such a direction as to **oppose the change of magnetic flux** that produces it. It is a statement of the conservation of energy.
1 (xii)	**Infrared** radiation.
1 (xiii)	Because of **total internal reflection**. Diamond has a very high refractive index (about 2·4), so its critical angle is only about 24°; light entering it is totally internally reflected many times before emerging.
1 (xiv)	The light from two independent sources has a **randomly varying phase difference** (each changes phase about 10^8 times a second), so the sources are not **coherent** and no steady pattern can persist.

Section B

Q	Answer
2	(i) $\tau = pE \sin 30° = (4 \times 10^{-9})(5 \times 10^4)(0{\cdot}5) = \mathbf{1{\cdot}0 \times 10^{-4}\ N\ m}$. (ii) $W = U_{90°} - U_{30°} = pE \cos 30° = (4 \times 10^{-9})(5 \times 10^4)(0{\cdot}866) = \mathbf{1{\cdot}73 \times 10^{-4}\ J}$.
3	$I = 2{\cdot}0/(3{\cdot}5+0{\cdot}50) = \mathbf{0{\cdot}50\ A}$; $V = \varepsilon - Ir = 2{\cdot}0 - (0{\cdot}50)(0{\cdot}50) = \mathbf{1{\cdot}75\ V}$ ($= IR$).
3 (OR)	Any two of: its **length** ($R \propto l$), its **area of cross-section** ($R \propto 1/A$), the **material** (through ρ), and its **temperature**. For a metal the resistivity **increases** with temperature: the lattice ions vibrate more, so the relaxation time τ falls, and $\rho = m/ne^2\tau$ rises.
4	$B = \mu_0 NI/2R = (4\pi \times 10^{-7})(20)(5{\cdot}0)/(2 \times 0{\cdot}080) = \mathbf{7{\cdot}9 \times 10^{-4}\ T}$, directed along the axis by the right-hand rule.
5	$\lvert\varepsilon\rvert = L(dI/dt) = (2{\cdot}0)(3{\cdot}0/0{\cdot}30) = \mathbf{20\ V}$.
6	Any two: they consist of mutually perpendicular oscillating electric and magnetic fields; they are **transverse**, with **E**, **B** and the direction of travel mutually perpendicular; they need no material medium; all travel at $c = 3 \times 10^8$ m s^{-1} in vacuum; they carry no charge and so are undeflected by electric and magnetic fields; they carry energy and momentum.

Q	Answer
6 (OR)	X-rays < ultraviolet < infrared < microwaves. **X-rays** have the greatest frequency, since frequency and wavelength run in opposite orders.
7	A curve which falls, passes through a **single minimum** and then rises, with i on the x-axis and δ on the y-axis; the minimum marked δ_m and the corresponding abscissa marked $i = e$. [1 for the correct shape and labelled axes, 1 for marking δ_m.] Note that for every δ other than δ_m there are **two** angles of incidence.
8	$\lambda = \beta d/D = (3{\cdot}0 \times 10^{-3})(0{\cdot}30 \times 10^{-3})/1{\cdot}5 = 6{\cdot}0 \times 10^{-7}$ m = **600 nm**.

Section C

Q	Answer
9	Each charge is a distance $\sqrt{(r^2+l^2)}$ from P, so $E_1 = E_2 = kq/(r^2+l^2)$. **[1]** Resolving: the components perpendicular to the axis are equal and opposite and **cancel**; those parallel to it **add**. **[1]** With $\cos\theta = l/\sqrt{(r^2+l^2)}$, $E = 2E_1 \cos\theta = \mathbf{kp/(r^2+l^2)^{3/2}}$, directed **antiparallel** to **p**. **[1]** For a short dipole $E = kp/r^3$, which is **half** the axial value at the same distance.
10	Diagram: driver cell ε_0 with key and rheostat in series with the potentiometer wire AB; the experimental cell (ε, r) connected through a galvanometer to the jockey; and a resistance box R with a key **across** the experimental cell. **[2]** $\mathbf{r = R[l_1/l_2 - 1]}$, where l_1 is the balancing length with the key **open** (corresponding to the emf), l_2 that with the key **closed** (corresponding to the terminal p.d.), and R the known shunt. **[1]**
11	The two sides of length l are perpendicular to **B**, so each carries a force $F = BIl$. They are equal, opposite and **not collinear**, so they form a **couple**; the forces on the other two sides are collinear and cancel. **[1]** The perpendicular distance between them is $b \sin\theta$, so $\tau = BIl{\cdot}b \sin\theta = BIA \sin\theta$, and for N turns $\boldsymbol{\tau} = \mathbf{NIAB \sin\theta}$. **[1]** Comparing with $\tau = mB \sin\theta$, the **magnetic dipole moment** is $\mathbf{m = NIA}$ (unit A m^2), normal to the plane of the loop; vectorially $\boldsymbol{\tau} = \mathbf{m} \times \mathbf{B}$. **[1]**
11 (OR)	Ampere's law: $\oint \mathbf{B}{\cdot}d\mathbf{l} = \mu_0 I$. Choose a **circle of radius r** in the plane perpendicular to the wire and centred on it; by symmetry B is constant in magnitude and **tangential**, so $\mathbf{B} \parallel d\mathbf{l}$ everywhere. **[1]** Then $\oint B\, dl = B{\cdot}2\pi r = \mu_0 I$. **[1]** Hence $\mathbf{B = \mu_0 I/2\pi r}$, so $B \propto 1/r$; the lines are concentric circles. **[1]**
12	$eV = \frac{1}{2}mv^2$ gives $v = \sqrt{(2eV/m)} = \sqrt{(2(1{\cdot}6 \times 10^{-19})(500)/9{\cdot}1 \times 10^{-31})} = 1{\cdot}33 \times 10^7$ m s^{-1}. Then $r = mv/eB = (9{\cdot}1 \times 10^{-31})(1{\cdot}33 \times 10^7)/((1{\cdot}6 \times 10^{-19})(2{\cdot}0 \times 10^{-3}))$ = $\mathbf{3{\cdot}8 \times 10^{-2}}$ **m = 3·8 cm**.
13	**Principle:** mutual induction. **[1] Working:** an alternating current in the primary produces a continually changing flux in the laminated soft-iron core; the same flux links the secondary and induces an alternating emf in it. **[1]** Since the same flux links both, $\mathbf{V_s/V_p = N_s/N_p}$, and for an ideal transformer $I_s/I_p = N_p/N_s$. **[1]** Efficiency is below 100% because of eddy-current, hysteresis, copper (I^2R) and flux-leakage losses (any one).
14	Diagram: plane wavefront AB incident obliquely, A reaching the surface at O while B travels on to O′ in time t; i and r marked; at least one direction arrow. In that time OC = BO′ = ct, both in the same medium. **[1]** In the right-angled triangles OBO′ and OCO′: OO′ is common, BO′ = OC = ct, and the angles at B and C are both 90°, so the triangles are **congruent**. **[1]** Hence ∠BOO′ = ∠CO′O, i.e. **i = r**, and all three of the incident ray, normal and reflected ray lie in one plane. **[1]**
15	At the first surface (radius R_1): $n_2/v' - n_1/u = (n_2-n_1)/R_1$. **[1]** That image is the **object** for the second surface (radius R_2), where light returns to n_1: $n_1/v - n_2/v' = (n_1-n_2)/R_2$. **[1]** Adding, v' cancels; dividing by n_1 and putting $u = \infty$, $v = f$ gives $\mathbf{1/f = (n-1)[1/R_1 - 1/R_2]}$ with $n = n_2/n_1$. **[1]**
16	$\sin C = 1/1{\cdot}5 = 0{\cdot}667$, so **C = 41·8°**. **[1]** Entering normally, the ray goes straight in and meets the hypotenuse at **45°**; since 45° > 41·8° it suffers **total internal reflection**, is turned through a right angle, and leaves the second face normally. **[1] Total deviation = 90°. [1]**
16 (OR)	$M = (f_o/f_e)(1 + f_e/D) = (100/5{\cdot}0)(1 + 5{\cdot}0/25) = 20 \times 1{\cdot}2 =$ **24**. In normal adjustment it would be only 20, so the magnifying power is greater with the image at D.
17	(i) The angular width $2\lambda/a$ is **halved**: the pattern narrows and the central maximum becomes brighter, since more light gets through and is concentrated into a smaller patch. (ii) The angular width is **doubled**, since it varies as λ. (iii) The **angular** width is unchanged, but the **linear** width $2\lambda D/a$ grows in proportion to D, so the pattern becomes larger but fainter.

Section D

Q	Answer
18	**(a)** X_L = 2π(50)(0·40) = 125·7 Ω; X_C = 1/[2π(50)(40 × 10^{-6})] = 79·6 Ω; $X_L - X_C$ = 46·1 Ω (net inductive, so the current lags); $Z = \sqrt{(40^2+46\cdot1^2)}$ = **61·0 Ω**; I = 220/61·0 = **3·6 A**. **(b)** A curve high at low f, falling to a **minimum Z = R** at f_0, then rising again; both f_0 and the minimum value R marked.
18 (OR)	**(a)** In time dt the rod sweeps area $lv\ dt$, so $d\Phi = Blv\ dt$ and **$\lvert\varepsilon\rvert$ = Blv**. **[1]** The induced current is Blv/R, and the force on the rod is $BIl = B^2l^2v/R$, **opposing the motion** (Lenz). **[1]** So the agent must supply **$P = Fv = (Blv)^2/R = I^2R$** – exactly the electrical power dissipated, so energy is conserved. **[1] (b)** ε = (0·20)(0·50)(4·0) = 0·40 V; I = **0·20 A**; $P = I^2R$ = **0·080 W**.
19	**(a)** Diagram: small object just beyond the objective's focus; real, inverted, magnified intermediate image just inside the eyepiece's focus; final virtual inverted magnified image at D on the object side; f_o, f_e, v_o, u_e and D labelled. **[2]** **$M = (v_o/u_o)(1 + D/f_e)$**. **[1] (b)** Objective: $1/v_o$ = 1/1·0 − 1/1·1 = 0·0909, so v_o = 11 cm and m_o = 11/1·1 = 10. Image at infinity, so $m_e = D/f_e$ = 25/5·0 = 5. **M = 50**.
19 (OR)	**(a)** Ray diagram with the object between the lens and its focus, giving a virtual erect magnified image at D; visual angles α and β marked. **[1]** $M = \beta/\alpha \approx (h/\lvert u\rvert)/(h/D) = D/\lvert u\rvert$. **[1]** With $v = -D$ in the lens formula, $1/\lvert u\rvert = 1/f + 1/D$, so **$M = 1 + D/f$**. **[1] (b)** Using $M = D/f$ for the relaxed-eye form, f = 25/10 = **2·5 cm** = 0·025 m, so P = **40 D**. (With $M = 1 + D/f$ instead, f = 25/9 = 2·8 cm; either is accepted if the formula used is stated.)
20	(i) $C = \varepsilon_0 A/d$. (ii) C = (8·85 × 10^{-12})(4·0 × 10^6)/2000 = **1·77 × 10^{-8} F = 17·7 nF**. (iii) $U = ½CV^2$ = ½(1·77 × 10^{-8})(10^{16}) = **8·85 × 10^7 J**. (iv) $Q = CV$ = 1·77 C, so $I = Q/t$ = 1·77/(1·0 × 10^{-4}) = **1·77 × 10^4 A** (about 17·7 kA). (v) Because the stroke carries an enormous current for a very short time. A thick strip has a very **low resistance**, so the heat generated (I^2Rt) is not enough to melt or vaporise it, and the charge reaches earth safely instead of passing through the building.

Mark distribution over the First Term chapters

Chapter	A	B	C	D	Total
Electrostatics	2	2	3	5	**12**
Current Electricity	2	2	3	—	**7**
Magnetic Effect of Current and Magnetism	3	2	6	—	**11**
Electromagnetic Induction and Alternating Current	2	2	3	5	**12**
Electromagnetic Waves	1	2	—	—	**3**
Optics — Ray Optics and Optical Instruments	2	2	6	5	**15**
Optics — Wave Optics	2	2	6	—	**10**
Total	**14**	**14**	**27**	**15**	**70**